

1.) Injective: Suppose $f(a_1) = f(a_2)$ for some $a_1, a_2 \in A$. Then

$$\begin{array}{ccc} (g \circ f)(a_1) & = & (g \circ f)(a_2) \\ \parallel & & \parallel \\ a_1 & & a_2 \end{array}$$

So $a_1 = a_2$.

Surjective: Fix $b \in B$. Set $a = g(b)$. Then $a \in A$ and

$$f(a) = f(g(b)) = b. \quad \square$$

2.) (a.) Prove: Fix $\varepsilon > 0$. As $\lim a_n = a$, $\exists N$ s.t.
 $|a_n - a| < \varepsilon \quad \forall n \geq N$.

In particular,

$$a = a_n + (a - a_n) \geq m - \varepsilon$$

Therefore $a \geq m - \varepsilon \quad \forall \varepsilon > 0$, and so $a \geq m$. $\square$

(b.) Disprove: Consider $a_n = \frac{1}{n}$ and $m = 0$. Then

$$\bullet a_n = \frac{1}{n} > 0 = m \quad \forall n \geq 1.$$

$$\bullet \lim_{n \rightarrow \infty} a_n = 0 \neq 0 = m \quad \square$$

3.) (a.) Base case: For $n=1$, $\frac{1}{1 \cdot 4} = \frac{1}{3 \cdot 1 + 1}$ is true.

Inductive step: Suppose that

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

for some $n \geq 1$. Then

$$\begin{aligned}
\frac{1}{1 \cdot 4} + \dots + \frac{1}{(3n-2)(3n+1)} + \frac{1}{(3n+1)(3n+4)} &= \frac{n}{3n+1} + \frac{1}{(3n+1)(3n+4)} \\
&= \frac{n(3n+4) + 1}{(3n+1)(3n+4)} \\
&= \frac{3n^2 + 4n + 1}{(3n+1)(3n+4)} \\
&= \frac{\cancel{(3n+1)}(n+1)}{\cancel{(3n+1)}(3n+4)} \\
&= \frac{n+1}{3n+4}
\end{aligned}$$

So the formula holds for $n+1$.

Together, we conclude that the statement is true $\forall n \in \mathbb{N}$ by induction. $\square$

(b.) By part (a.),

$$S_n = \sum_{k=1}^n \frac{1}{(3k-2)(3k+1)} = \frac{n}{3n+1}$$

By the limit laws,

$$\frac{n}{3n+1} = \frac{1}{3 + \frac{1}{n}} \rightarrow \frac{1}{3} \quad \text{as } n \rightarrow \infty$$

Therefore $\lim S_n = \frac{1}{3}$, and so the series converges to $\frac{1}{3}$. $\square$

4.) We'll use the Cauchy criterion. Fix $\varepsilon > 0$.

As $\sum |a_n|$ converges, $\exists N$ s.t.

$$\sum_{k=m}^n |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

So, for $n \geq m \geq N$ we have

$$\left| \sum_{k=m}^n a_{2k} \right| \leq \sum_{k=m}^n |a_{2k}| \leq \sum_{l=2m}^{2n} |a_l| < \varepsilon.$$

Therefore $\sum a_{2n}$ converges by the Cauchy criterion. $\square$

5.) (a.) By lecture, it suffices to prove that Y is open in $\mathbb{R}^2$. Fix $(x, y) \in Y$. Then $y > 0$.

Set $r = y$. Then $r > 0$ and $B_r^{\mathbb{R}^2}(x, y) \subseteq Y$, since

$$\begin{aligned} (\tilde{x}, \tilde{y}) \in B_r^{\mathbb{R}^2}(x, y) &\Rightarrow \sqrt{|x - \tilde{x}|^2 + |y - \tilde{y}|^2} < r \\ &\Rightarrow |y - \tilde{y}| \leq \sqrt{|x - \tilde{x}|^2 + |y - \tilde{y}|^2} < r \\ &\Rightarrow \tilde{y} = y + (\tilde{y} - y) > y - r = 0 \\ &\Rightarrow (\tilde{x}, \tilde{y}) \in Y. \quad \square \end{aligned}$$

(b.) Consider

$$B = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 < y < 1\}.$$

Then B is closed in Y , since $B = [0, 1]^2 \cap Y$ and $[0, 1]^2$ is closed in $\mathbb{R}^2$. But B is not closed in $\mathbb{R}^2$, since $(0, 0) \notin B$ and $(0, 0) \in \overline{B}^{\mathbb{R}^2}$:

$$r > 0 \Rightarrow (0, \min\{\frac{r}{2}, 1\}) \in B \cap B_r^{\mathbb{R}^2}((0, 0)). \quad \square$$

6.) (a.) Prove: Fix $A \subseteq \mathbb{R}$ and $y \in f(\overline{A})$. Want: $y \in \overline{f(A)}$.

As $y \in f(\overline{A})$, then $\exists x \in \overline{A}$ s.t. $f(x) = y$. As $x \in \overline{A}$, then $\exists$ a sequence $\{x_n\}$ in A s.t. $x_n \rightarrow x$ as $n \rightarrow \infty$. As f is continuous, then $f(x_n) \rightarrow f(x) = y$ as $n \rightarrow \infty$.

Note that $f(x_n) \in f(A) \forall n$, and so $y \in \overline{f(A)}$. $\square$

(b.) Disprove: Consider $f(x) = \frac{1}{1+x^2}$ and $A = [0, \infty)$.
Note that $\overline{A} = A$ and $f(A) = (0, 1]$. So

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}. \quad \square$$

7.) Fix $F \subseteq X$ closed. As X is compact, then F is compact as well. As f is continuous, then $f(F)$ is compact in Y , and so is closed in Y .
Finally, note that $f(F) = (f^{-1})^{-1}(F)$ since f is bijective. $\square$

8.) (a.) Set $a_n = \sqrt[n]{n}$. Then

$$|a_n|^{\frac{1}{n}} = n^{\frac{1}{2n}} = (n^{\frac{1}{n}})^{\frac{1}{2}} \rightarrow (1)^{\frac{1}{2}} = 1 \text{ as } n \rightarrow \infty$$

and so

$$R = \frac{1}{\limsup |a_n|^{\frac{1}{n}}} = 1.$$

(b.) By part (a.), we know the series converges if $|x| < 1$ and diverges if $|x| > 1$.

- At $x = 1$: $\sum \sqrt[n]{n}$ diverges, since $\sqrt[n]{n} \not\rightarrow 0$ as $n \rightarrow \infty$.
- At $x = -1$: $\sum (-1)^n \sqrt[n]{n}$ diverges, since $(-1)^n \sqrt[n]{n} \not\rightarrow 0$ as $n \rightarrow \infty$.

So the series converges if and only if $x \in (-1, 1)$. $\square$